Lecture 19

Matrices and composition

Uses.

Summary 2: A composition applies the rightmost function first. Equality of functions means equality on every input.
Summary 12: Columns have componentwise addition and scalar multiplication. The standard coordinate columns are independent and span the coordinate space.
Summary 15: Complex conjugation preserves sums and products. The local 2×2 product represents multiplication of complex numbers.
Summary 17: A matrix has a specified number of rows and columns. Its action on a column records a linear system. Elementary row operations preserve the solutions of a system.
Summary 18: The coefficient action is linear. A basis gives unique coordinates, and the null space consists of the columns sent to zero.

Objectives. *A matrix describes how a linear map acts on coordinate columns. Applying two such maps in succession determines their matrix product. This interpretation proves the product laws, fixes the order of every operation and expresses row operations by multiplication.*

By the end of this lecture you will be able to:

- Construct a matrix from its action on the standard coordinate columns.
- Compute compatible products and prove their composition and associativity laws.
- Prove the transpose, adjoint and trace identities with their size restrictions.
- Represent row operations by elementary matrices and verify a two-sided inverse.

19.1 Matrix operations and columns

Keep the scalar field $K = \mathbb{R}$ or $K = \mathbb{C}$ fixed. For positive integers m and n, write $K^{m\times n}$ for the matrices with m rows and n columns and entries in K. Two matrices are equal if their sizes agree and their corresponding entries agree. For $A, B \in K^{m\times n}$ and $c \in K$, define addition and scalar multiplication by

$$(A + B)_{ij} = a_{ij} + b_{ij}, \qquad (cA)_{ij} = ca_{ij}, \qquad 1 \le i \le m, \quad 1 \le j \le n.$$

The zero matrix $0_{m\times n}$ has every entry zero. These entrywise operations inherit the associative and distributive scalar laws. Addition is commutative and $-A$ is the additive inverse of A, since each corresponding scalar assertion holds in every entry. Addition of matrices of different sizes is undefined.

Denote the columns of A by $a_1, \dots, a_n \in K^m$. The coefficient action from Lecture 17 has the column expression

$$Ax = x_1a_1 + \cdots + x_na_n, \qquad x \in K^n. \tag{19.1}$$

Indeed, its ith entry is $\sum_j a_{ij}x_j$, the previously defined action. For the standard coordinate column $e_j \in K^n$, only coordinate j is 1. Consequently $Ae_j = a_j$: a matrix's action on e_j recovers its jth column. The standard columns in different coordinate spaces have their sizes supplied by the expression in which they occur.

A map $T : K^n \to K^m$ is *linear over* K if

$$T(cu + dv) = cT(u) + dT(v) \qquad (u, v \in K^n,\ c, d \in K).$$

The coefficient action $x \mapsto Ax$ is linear by Eq. (18.2). Conversely, every linear map has exactly one matrix in the standard coordinates. To prove existence, use $x = \sum_j x_je_j$ and linearity repeatedly to obtain

$$T(x) = \sum_{j=1}^{n} x_jT(e_j).$$

Thus the matrix with columns $T(e_1), \dots, T(e_n)$ has action T by Eq. (19.1). For uniqueness, any matrix with that action must have these columns, since evaluating its action at each e_j recovers them. In particular, matrices of the same size with equal actions on every column are equal.

Example 19.1 (Recovering columns from nonstandard inputs)**.** Suppose $T : \mathbb{R}^2 \to \mathbb{R}^2$ is linear and

$$T\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}3\\4\end{pmatrix}, \qquad T\begin{pmatrix}1\\2\end{pmatrix} = \begin{pmatrix}4\\7\end{pmatrix}.$$

Neither supplied input is a standard coordinate column. Put $u = (1, 1)^T$ and $v = (1, 2)^T$. Since $e_1 = 2u - v$ and $e_2 = v - u$, linearity gives

$$T(e_1) = 2\begin{pmatrix}3\\4\end{pmatrix} - \begin{pmatrix}4\\7\end{pmatrix} = \begin{pmatrix}2\\1\end{pmatrix}, \qquad T(e_2) = \begin{pmatrix}4\\7\end{pmatrix} - \begin{pmatrix}3\\4\end{pmatrix} = \begin{pmatrix}1\\3\end{pmatrix}.$$

These are the columns of the unique standard matrix $A = \begin{pmatrix}2 & 1\\1 & 3\end{pmatrix}$. Checking $Au = T(u)$ and $Av = T(v)$ recovers the given data. The key step was that u and v span $\mathbb{R}^2$, so their images determine the image of every input. If supplied inputs do not span the domain, their images need not determine a unique map; if they are dependent, any proposed images must also obey the same linear relations for a linear map to exist.

19.2 Multiplication from successive actions

Let $A \in K^{m\times n}$ and $B \in K^{n\times p}$. The map $x \mapsto A(Bx)$ sends K^p to K^m and is linear: linearity of B and then of A takes a linear combination to the same combination of the two output columns. Define the *product* $AB \in K^{m\times p}$ to be its matrix. The defining action and entry formula are

$$(AB)x = A(Bx), \qquad (AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}. \tag{19.2}$$

To derive the entries, apply the composition to $e_j \in K^p$. The column Be_j is the jth column of B. Its image under A has ith entry $\sum_k a_{ik}b_{kj}$, which is the required entry of the jth product

column. Thus the inner sizes must agree. The outer sizes give the product's size. This definition extends the local 2×2 product of Lecture 15.

For a rectangular example, take

$$A = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 0 \\ 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

The product AB is 2×2. Taking each row against each column gives

$$AB = \begin{pmatrix} -1 & 5 \\ 2 & -1 \end{pmatrix}.$$

In the other order, the product is 3×3:

$$BA = \begin{pmatrix} 2 & -2 & 4 \\ 1 & -4 & 1 \\ -1 & 7 & 0 \end{pmatrix}.$$

For example, its middle row is the first row of A minus the second, namely $(1, -1, 2) - (0, 3, 1) = (1, -4, 1)$.

Product laws

The *identity matrix* $I_n \in K^{n \times n}$ has entries 1 on the diagonal and 0 elsewhere. Its action is $I_n x = x$. Hence $I_m A = A = A I_n$ for $A \in K^{m \times n}$, since both sides have the same action on every column. For compatible matrices, multiplication is associative:

$$(AB)C = A(BC), \qquad A \in K^{m \times n}, \quad B \in K^{n \times p}, \quad C \in K^{p \times q}. \tag{19.3}$$

For $x \in K^q$, the left action is $A(B(Cx))$ by two applications of the product definition. The right action has exactly the same value. Equality of actions proves equality of the matrices.

The distributive laws and the scalar rule, whenever the indicated products and sums have matching sizes, are

$$A(B + C) = AB + AC, \qquad (A + B)C = AC + BC, \qquad (cA)B = c(AB) = A(cB).$$

For the first law, linearity of A gives $A((B + C)x) = A(Bx + Cx) = A(Bx) + A(Cx)$. For the second, the entrywise sum acts by $(A + B)(Cx) = A(Cx) + B(Cx)$. For the scalar rule, scalar multiplication of an action and linearity give $(cA)(Bx) = cA(Bx) = A(cBx)$. Each proves the corresponding equality on all inputs, and therefore on matrices.

For a square matrix $A \in K^{n \times n}$, define its powers by $A^0 = I_n$ and $A^{r+1} = A^r A$ for integers $r \geq 0$. Associativity gives $A^r A^s = A^{r+s}$ for $r, s \geq 0$: induction on s starts with I_n, and multiplying the induction equality by A gives the next case. These powers are defined only for square matrices.

19.3 Order of multiplication and plane motions

Associativity does not permit changing the order of factors. For $\theta \in \mathbb{R}$, the counterclockwise plane rotation has matrix

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Indeed, its columns are the rotated standard unit columns, by the angle addition formulas of Standing Fact E. Those same formulas in the four product entries give $R_\theta R_\phi = R_{\theta+\phi}$ for $\theta, \phi \in \mathbb{R}$. For example, the upper-left entry is $\cos\theta\cos\phi - \sin\theta\sin\phi = \cos(\theta+\phi)$; the upper-right is $-\sin(\theta+\phi)$, the lower-left is $\sin(\theta+\phi)$, and the lower-right is $\cos(\theta+\phi)$.

The *horizontal shear* S sends (x, y) to $(x+y, y)$. Its matrix and a quarter-turn matrix are

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad R = R_{\pi/2} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Figure 19.1 compares their successive actions on e_1.

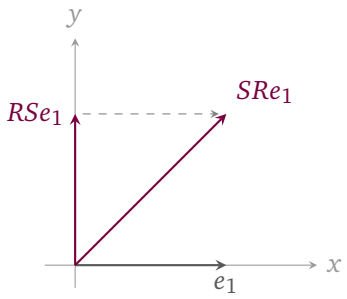


Figure 19.1: The shear fixes e_1, so a following quarter-turn gives $(0, 1)$. Reversing the order shears the rotated column to $(1, 1)$.

The products are

$$RS = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}, \qquad SR = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}.$$

Their first columns already differ. Therefore $RS \neq SR$, although both orders have the same size.

A zero product also need not have a zero factor. For example, the nonzero matrices

$$E = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

have $EF = 0_{2\times 2}$. The action of F keeps only the second coordinate, which E sends to zero. Thus a product can vanish because the first action's outputs all lie in the null space of the following action.

19.4 Transpose, adjoint and trace

For $A \in K^{m\times n}$, its *transpose* $A^{\mathsf{T}} \in K^{n\times m}$ is given by $(A^{\mathsf{T}})_{ji} = a_{ij}$. For a complex matrix, its entrywise conjugate $\overline{A}$ has entries $\overline{a_{ij}}$. Its *adjoint* is the conjugate transpose

$$A^\dagger = \overline{A}^{\mathsf{T}}, \qquad (A^\dagger)_{ji} = \overline{a_{ij}}.$$

For real matrices the adjoint equals the transpose. For example,

$$C = \begin{pmatrix} 1+\mathrm{i} & 2 \\ -\mathrm{i} & 3-2\mathrm{i} \end{pmatrix}, \qquad C^\dagger = \begin{pmatrix} 1-\mathrm{i} & \mathrm{i} \\ 2 & 3+2\mathrm{i} \end{pmatrix}.$$

The two operations of conjugation and transposition have different roles: conjugation changes entries, while transposition interchanges their positions.

For $A \in K^{m\times n}$ and $B \in K^{n\times p}$, the product rules are

$$(AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}}, \qquad (AB)^{\dagger} = B^{\dagger}A^{\dagger}. \tag{19.4}$$

Both right sides have size $p \times m$. For the transpose, their entry in row j, column i is

$$\left((AB)^{\mathsf{T}}\right)_{ji} = \sum_{k=1}^{n} a_{ik}b_{kj} = \sum_{k=1}^{n} (B^{\mathsf{T}})_{jk}(A^{\mathsf{T}})_{ki}.$$

For the adjoint, conjugate the same finite sum. Conjugation preserves addition and products by Lecture 15, giving

$$\overline{\sum_{k=1}^{n} a_{ik}b_{kj}} = \sum_{k=1}^{n} \overline{b_{kj}}\,\overline{a_{ik}} = (B^{\dagger}A^{\dagger})_{ji}.$$

These entry identities prove both product rules. The definitions also give $(A^{\mathsf{T}})^{\mathsf{T}} = A$ and $(A^{\dagger})^{\dagger} = A$, since each entry returns to its original position and conjugation twice returns the original scalar. For same-size matrices, transposition preserves sums and scalar multiples; the adjoint preserves sums but sends cA to $\overline{c}A^{\dagger}$. Each assertion follows by applying its entry definition to the sum or multiple.

For a square matrix $D \in K^{n\times n}$, its *trace* is the diagonal sum

$$\operatorname{tr} D = \sum_{i=1}^{n} d_{ii}.$$

The trace of a rectangular matrix is not defined here. For $A \in K^{m\times n}$ and $B \in K^{n\times m}$, both products are square, and the finite double sum proves

$$\operatorname{tr}(AB) = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}b_{ji} = \sum_{j=1}^{n}\sum_{i=1}^{m} b_{ji}a_{ij} = \operatorname{tr}(BA). \tag{19.5}$$

The middle equality interchanges finite sums and commutes scalar factors. It does not commute the matrices. In the rectangular example above, the traces are $-1 - 1 = -2$ and $2 - 4 + 0 = -2$, even though the products have different sizes.

19.5 Elementary matrices and inverses

An *elementary matrix* of size m is obtained by performing one of Lecture 17's elementary row operations on I_m. If this operation exchanges rows i and j, the resulting matrix has rows e_j^{T} and e_i^{T} in those positions. Its product with any $A \in K^{m\times n}$ therefore exchanges those rows of A. If the operation multiplies row i by $c \neq 0$, the corresponding row of the elementary matrix is ce_i^{T}; its product row is c times row i of A. If the operation adds c times row j to a different row i, its row i is $e_i^{\mathsf{T}} + ce_j^{\mathsf{T}}$; its product row is row i of A plus c times row j. All other rows equal the corresponding coordinate rows and leave A unchanged. This proves that each elementary row operation is left multiplication by its elementary matrix.

For the earlier 2×3 matrix A, adding three times row 1 to row 2 uses the elementary matrix T. The matrix and its resulting product are

$$T = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}, \qquad TA = \begin{pmatrix} 1 & -1 & 2 \\ 3 & 0 & 7 \end{pmatrix}.$$

The new second row is $3(1,-1,2)+(0,3,1)=(3,0,7)$. For a sequence of operations with elementary matrices $E_1,\dots,E_r$, the final array is $E_r\cdots E_1A$. The first operation is nearest to A because it acts first. The same statement applies to the whole augmented array $[A \mid b]$.

A square matrix $A \in K^{n\times n}$ is *invertible* if there exists a square matrix C satisfying both identities

$$AC = CA = I_n. \tag{19.6}$$

Such a matrix is its *inverse*, denoted by A^{-1}. The inverse is unique. If C and D both satisfy the two identities, associativity gives $C = C(AD) = (CA)D = D$. Each elementary matrix has an inverse: take the elementary matrix of its reverse row operation from Lecture 17. Performing the operation and then its reverse returns every matrix, in particular I_m, to itself. Reversing that order also does so, proving both products equal I_m. For the displayed row-addition matrix T, the reverse operation subtracts three times row 1 from row 2.

For rotations the previously proved product formula gives $R_\theta R_{-\theta} = R_{-\theta}R_\theta = R_0 = I_2$, so $R_\theta^{-1} = R_{-\theta}$. Lecture 20 gives a general procedure to decide invertibility by elimination.

Summary 19.

- Equation (19.1): Ax combines the columns of A with coefficients from x. A linear map's unique standard matrix has columns $T(e_j)$.
- Equations (19.2) and (19.3): compatible matrix multiplication represents composition and is associative. Factor order cannot generally be changed.
- Equations (19.4) and (19.5): transpose and adjoint reverse product order. The trace identity holds for $A \in K^{m\times n}$ and $B \in K^{n\times m}$.
- Elementary matrices perform row operations by left multiplication. A sequence of operations has product $E_r\cdots E_1$ in the order of action.
- Equation (19.6): an inverse satisfies both identities and is unique. Elementary matrices are invertible. The rotation inverse is $R_{-\theta}$.

Sheet 19 practices products, powers, traces and elementary operations. Lecture 20 uses these operations to construct an inverse.

References

D. C. Lay, S. R. Lay and J. J. McDonald, *Linear Algebra and Its Applications*, 5th ed., Pearson, 2016, Chapter 2, §§2.1–2.2. S. Lang, *Introduction to Linear Algebra*, 2nd ed., Springer, 1986, Chapter II, §§1–3.